KCG COLLEGE OF TECHNOLOGY
AFFILIATED TO ANNA UNIVERSITY | AUTONOMOUS

Unit I

	CSE(Cyber Security)		
	23CB402 – Cyber Security		
	First		
QNo	**Questions**	**COs**	**Bloom's Level**
1.	Define cyber security.	CO1	K1
2.	List the elements of cyber security.	CO1	K1
3.	Define Cyber Crime	CO1	K1
4.	Illustrate CIA triad with a simple diagram.	CO1	K2
5.	List out the positive and negative impacts of internet in society.	CO1	K1
6.	Interpret the reasons for cyber crime.	CO1	K2
7.	Define DNS	CO1	K1
8.	Summarize espionage.	CO1	K2
9.	List the types of cyber criminals.	CO1	K1
10.	Compare Black hat, grey hat and white hat hackers	CO1	K2
11.	Classify cyber crimes.	CO1	K2
12.	What is cyber crime against individual?Give example.	CO1	K2
13.	What is cyber crime against property in short. Give example.	CO1	K2
14.	What is cyber crime against organization. Give example.	CO1	K2
15.	What is cyber crime against society. Give example.	CO1	K2
16.	Define cyber law with an example.	CO1	K2
17.	Classify cyber law with its types.	CO1	K2
18.	Outline the offenses and punishments in Indian IT act.	CO1	K2
19.	Summarize the importance of cyber security.	CO1	K2
20.	What is ARPANET in the development of the internet?	CO1	K2
	Part – B		
1.	Explain cyber security and its types in detail.	CO1	K2
2.	Demostrate CIA triad with suitable examples	CO1	K2
3.	i.Enumerate cyber crime and its types in detail. (8) ii.List the need for cyber crime in detail.(8)	CO1	K2
4.	i.Outline history of internet in detail.(8)	CO1	K2

	ii.Illustrate the impact of internet in society.(8)		
5.	Demonstrate history of cyber crime in detail with examples	CO1	K2
6.	Summarize cyber criminals and classification of cyber crime in detail.	CO1	K2
7.	Illustrate Cyber law and its types in detail	CO1	K2
8.	Explain Indian IT act in detail with cyber crime and punishments.	CO1	K2

UNIT II

QNo	Questions	COs	Bloom's Level
1.	Define OWASP.	CO2	K1
2.	List out OWASP Top 10	CO2	K1
3.	How does a Trojan horse differ from other types of malicious software.	CO2	K1
4.	Summarize the scope of cyber attacks	CO2	K2
5.	Compare malicious attacks and threats.	CO2	K2
6.	Outline active threats with its types.	CO2	K2
7.	Describe security breach with a real time example	CO2	K2
8.	List out the types of malicious attacks	CO2	K1
9.	List the types of malwares with an example each.	CO2	K2
10.	What is malicious software. Give example.	CO2	K2
11.	Compare virus and worms	CO2	K2
12.	Outline common attack vectors in short.	CO2	K2
13.	What is social engineering attack? Give example.	CO2	K2
14.	What is wireless network attack? Give example.	CO2	K2
15.	What is web application attack? Give example.	CO2	K2
16.	Enumerate the types of wireless network attacks.	CO2	K2
17.	Outline the methods to prevent wireless network attacks.	CO2	K1
18.	List out the types of web application attacks	CO2	K1
19.	What are the web application security strategies that organizations can implement to protect their web applications	CO2	K2
20.	List the attack tools used by cyber criminals.	CO2	K2
Part – B			

1.	Develop a checklist for securing web applications using OWASP 10 guidelines.	CO2	K3
2.	Demonstrate security breach with its types and real time examples.	CO2	K2
3.	Outline the types of malwares in detail with examples	CO2	K2
4.	Identify the most common social engineering attacks and explain in detail how to prevent them.	CO2	K3
5.	Choose suitable hardware and software tools to discover exploitable weaknesses for the actual attack by cybercriminals	CO2	K3
6.	Identify the attack vectors for suitable attacks launched by cyber attackers	CO2	K3
7.	Analyse the attack that targets the weaknesses in the protocols, configurations, or encryption mechanisms of wireless networks and explain it in detail.	CO2	K4
8.	Analyse the attacks that can result in unauthorized access, data theft, or other harmful consequences and explain them in detail.	CO2	K4

UNIT III

QNo	Questions	COs	Bloom's Level
1.	Summarize the purpose of harvester in short.	CO3	K2
2.	Define Whois with an example.	CO3	K1
3.	List the key aspects of Netcrafts.	CO3	K1
4.	Mention the key tools for host reconnaissance	CO3	K1
5.	Summarize the key techniques in social engineering reconnaissance.	CO3	K2
6.	Classify the technique that involve leaving infected USB drives in public places or offering free downloads that contain malware.	CO3	K2
7.	Illustrate the steps taken by organisations to protect against social engineering attacks by reducing their susceptibility during the reconnaissance phase	CO3	K2
8.	Mention the types of scanning in reconnaissance.	CO4	K1
9.	Compare network scanning and vulnerability scanning.	CO4	K2
10.	Relate ping scan, SyNC scan and XMAS scan.	CO4	K2
11.	Interpret the categories of port scan that reveal the status of network or server.	CO4	K2
12.	Mention the purpose of network scanning.	CO4	K1

13.	Compare credentialed scan and non-credentialed scan.	CO4	K2
14.	Enumerate the steps for scanning methodologies.	CO4	K2
15.	How can banner grabbing be useful for identifying the services running on a target system?	CO4	K1
16.	List important flags in Nmap switches.	CO4	K1
17.	Relate NULL scan to XMAS and FIN	CO4	K2
18.	Mention the limitations of XMAS and NULL scan	CO4	K1
19.	How can organizations protect against banner grabbing to avoid revealing sensitive information?	CO4	K1
20.	List the important elements to determine the operating system in fingerprinting.	CO4	K1
Part – B			
1.	Explain the following i.Harvester ii.Whoi iii. Netcraft	CO3	K2
2.	Identify the methods used in reconnaissance and explain in detail.	CO3	K3
3.	Identify the process of gathering information about individuals, organizations, or systems through human interaction and manipulation, rather than through technical means and explain its types with examples.	CO3	K3
4.	Compare port scan, network and vulnerability scanning in detail.	CO3	K2
5.	Analyse the steps in scanning methodologies with examples.	CO4	K3
6.	Identify the commands used in Nmap and explain elaborately.	CO4	K3
7.	Relate Syn, XMAS, idle, NULL, FIN, stealth scans with its working in detail.	CO4	K2
8.	Compare banner grabbing and OS finger printing and explain with examples in detail.	CO4	K2

UNIT IV

QNo	Questions	Cos	Bloom's Level
1.	What is Intrusion Detection System?	CO5	K1
2.	Classify Intrusion Detection System.	CO5	K2
3.	Differentiate Network and Host intrusion detection system.	CO5	K2
4.	Mention the Detection Method of Intrusion Detection System.	CO5	K1

5.	List the categories of Host-Based Intrusion Detection Systems	CO5	K1
6.	What are the limitations and challenges of Host-Based Intrusion Detection Systems.	CO5	K1
7.	Classify the different detection methods to identify suspicious traffic and abnormal behaviour in network Intrusion Detection Systems	CO5	K2
8.	Compare the advantages and limitations of network Intrusion Detection Systems.	CO5	K2
9.	Abbreviate IDMEF.	CO5	K2
10.	Mention the key characteristics of Distributed Intrusion Detection Systems.	CO5	K1
11.	Relate the advantages and disadvantages of Distributed Intrusion Detection Systems.	CO5	K2
12.	Define honeypot with an example.	CO5	K1
13.	Classify honeypot based on deployment and interaction.	CO5	K2
14.	Differentiate advantages and disadvantages of honeypot.	CO5	K2
15.	Summarize SNORT briefly.	CO5	K2
16.	What Are the Different SNORT Modes?	CO5	K1
17.	Illustrate the Uses of SNORT Rules.	CO5	K2
18.	Differentiate Normal Internet Activities and Malicious Activities by SNORT	CO5	K2
19.	Mention the features of SNORT.	CO5	K1
20.	Classify honeypot by the type of activity they detect.	CO5	K2
Part – B			
1.	Classify Intrusion Detection methods and explain each of them briefly along with detection methods.	CO5	K2
2.	Examine how Host-Based Intrusion Detection (HIDS) interacts with the host operating system to identify malicious activities.	CO5	K3
3.	Identify the network traffic to detect intrusions by Network-Based Intrusion Detection System (NIDS) and explain in detail.	CO5	K3
4.	Apply the concept of Hybrid Intrusion Detection Systems (HIDS + NIDS) to enhance security over standalone systems.	CO5	K3
5.	Describe in detail the use of standardized formats like IDMEF to improve interoperability among intrusion detection systems.	CO5	K2
6.	Identify the various activities, deployment and interaction of honeypot and classify them in detail.	CO5	K3
7.	Explain the steps involved in configuring and deploying Snort in a network.	CO5	K2
8.	Compare SNORT with other intrusion detection systems in terms of performance and usability	CO5	K3

UNIT V

QNo	Questions	COs	Bloom's Level
1.	Illustrate the primary need for firewalls in network security.	CO6	K2
2.	Demonstrate the key characteristics of a firewall.	CO6	K2
3.	Describe access policy in the context of firewalls.	CO6	K2
4.	List the types of firewalls.	CO6	K1
5.	Interpret packet-filtering firewall and its functions.	CO6	K2
6.	Differentiate stateful inspection firewall from a stateless firewall.	CO6	K2
7.	Summarize application-level gateway firewall in short.	CO6	K2
8.	Define firewall basing with an example.	CO6	K1
9.	Compare host-based and network-based firewalls.	CO6	K2
10.	Interpret the location of firewall in a network for optimal security with a neat diagram.	CO6	K1
11.	Relate DMZ (Demilitarized Zone) to firewall configurations.	CO6	K2
12.	Illustrate a single-homed firewall setup with an example.	CO6	K2
13.	Infer how a dual-homed firewall enhance security.	CO6	K2
14.	What is the primary purpose of an Intrusion Prevention System (IPS)?	CO6	K1
15.	Relate Intrusion Prevention System (IPS) from Intrusion Detection System (IDS)	CO6	K2
16.	List the types of IPS based on deployment.	CO6	K1
17.	What is a host-based IPS (HIPS), and where is it deployed?	CO6	K1
18.	What is the role of a network-based IPS (NIPS)?	CO6	K1
19.	Outline the significance of Unified Threat Management (UTM).	CO6	K2
20.	Mention two example products that offer Unified Threat Management(UTM) solutions	CO6	K1
	Part – B		
1.	Explain the role of firewalls and their protection against external threats to ensure secure communication between trusted and untrusted networks.	CO6	K2
2.	Discuss the essential characteristics of a firewall to access policies influencing the functions and explain the key considerations defining these policies	CO6	K2
3.	Compare and contrast the different types of firewalls. Provide real time scenarios where each type is most effective.	CO6	K2
4.	Illustrate firewall basing, and how does it affect the deployment of	CO6	K2

	firewalls also describe the significance of firewall location and common configurations like single-homed, dual-homed, and DMZ setups.		
5.	i. Summarize Intrusion Prevention System (IPS) its working in detail.(10) Ii Differentiate Intrusion Prevention System(IPS) and Intrusion Detection System (IDS).(6)	CO6	K2
6.	Classify Intrusion Prevention System (IPS) and apply real time examples for each of it	CO6	K2
7.	Develop a UTM product to illustrate how it combines firewall, IPS, VPN, and anti-malware capabilities to protect against cyber threats efficiently in an organisation.	CO6	K3
8.	Apply network-based IPS to block malicious traffic in real-time while minimizing false positives.	CO6	K3